

Angular Hands-On 8

HTTP Client, Interceptors & Error Handling

Objectives

- Learn how to use Angular's HttpClient for API communication.
- Perform GET and POST requests to a REST API.
- Set up a mock backend using JSON Server.
- Implement HTTP Interceptors for logging and error handling.
- Display API data dynamically in components.

Tasks

Task 1: Setup Mock Backend

- Install JSON Server:

```
npm install -g json-server
```

- Create db.json with sample course data:

```
json
```

```
{
```

```
"courses": [  
  { "id": 1, "name": "Math", "code": "M101", "credits": 3 },  
  { "id": 2, "name": "Physics", "code": "P102", "credits": 4 }  
]  
}
```

- Run server:

```
json-server --watch db.json --port 3000
```

Task 2: Create API Service

- Generate service:

```
ng generate service services/api
```

- Implement methods in api.service.ts:
 - getCourses() → GET request
 - addCourse(course) → POST request

Task 3: Display API Data

- Inject ApiService into CourseListComponent.
- Call getCourses() in ngOnInit().
- Bind results in template with *ngFor.

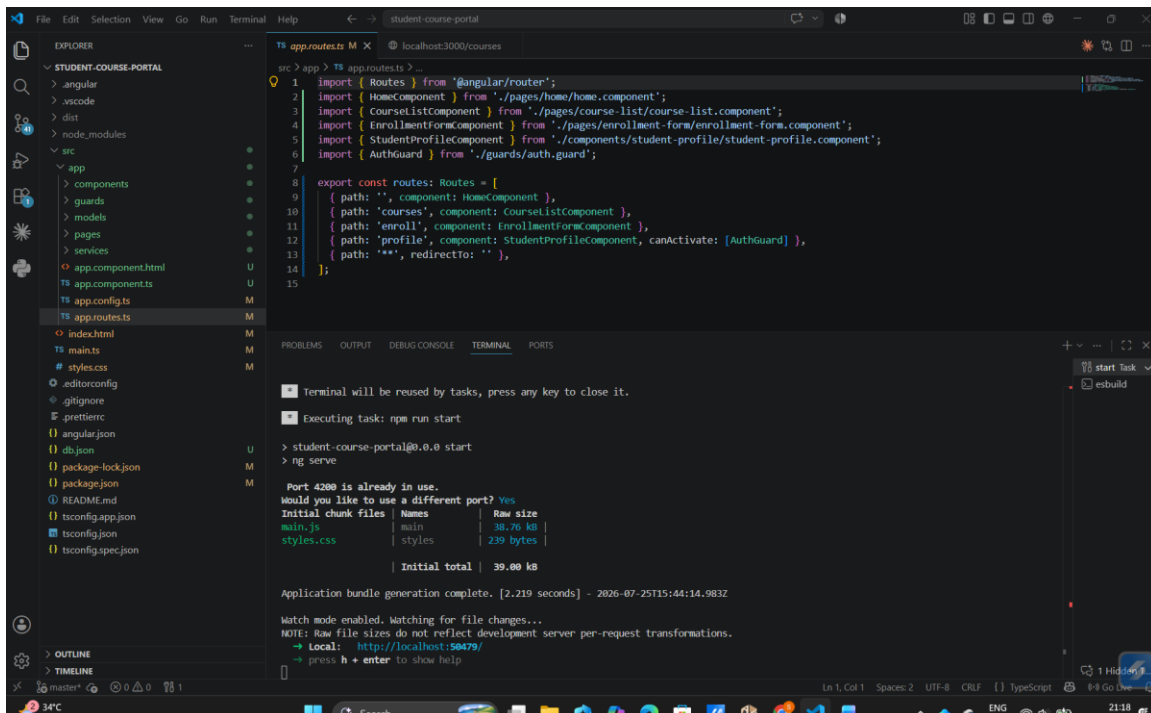
Expected Outcome: Courses load dynamically from API.

Task 4: Implement HTTP Interceptor

- Generate interceptor:

ng generate interceptor services/http-error

- In http-error.interceptor.ts:
 - Log all requests.
 - Catch errors with catchError.
 - Show alert message for errors.
- Register interceptor in app.module.ts providers.



The screenshot shows a Visual Studio Code editor with a project named 'student-course-portal'. The Explorer panel on the left shows the file structure, including 'src/app' and 'src/guards'. The main editor displays the file 'src/app/app.routes.ts' with the following code:

```
1 import { Routes } from '@angular/router';
2 import { HomeComponent } from './pages/home/home.component';
3 import { CourseListComponent } from './pages/course-list/course-list.component';
4 import { EnrollmentFormComponent } from './pages/enrollment-form/enrollment-form.component';
5 import { StudentProfileComponent } from './components/student-profile/student-profile.component';
6 import { AuthGuard } from './guards/auth.guard';
7
8 export const routes: Routes = [
9   { path: '', component: HomeComponent },
10  { path: 'courses', component: CourseListComponent },
11  { path: 'enroll', component: EnrollmentFormComponent },
12  { path: 'profile', component: StudentProfileComponent, canActivate: [AuthGuard] },
13  { path: '**', redirectTo: '' },
14 ];
15
```

The terminal panel at the bottom shows the output of the command 'ng generate interceptor services/http-error'. The output indicates that the interceptor was generated successfully and provides the following information:

```
Port 4200 is already in use.
Would you like to use a different port? Yes
Initial chunk files
main.js      38.76 kb
styles.css   239 bytes
Initial total 39.00 kb
Application bundle generation complete. [2.219 seconds] - 2026-07-25T15:44:14.983Z
Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:50479/
→ press h + enter to show help
```

